THESIS PROJECT UPDATE

Severe Legionnaires' disease in Wisconsin, 2023–2025

What changed in the data and code, what we're trying to do, and what to decide next

For Rachel Erickson · prepared by Johnny Uelmen, Uelmen Disease Ecology Lab, UW–Madison

November 2026 · ahead of the late-January committee meeting

THE PROJECT

What we're trying to answer, and why

PRIMARY AIM

Among the 635 confirmed Wisconsin LD cases (2023–2025), are reported exposures — healthcare, senior living, recreational water, travel — linked to more severe outcomes once age, sex, race/ethnicity and underlying conditions are accounted for?

SECONDARY AIM (time and data permitting)

Describe where and when cases occur: population- and age-adjusted maps, seasonality, and area-level environmental and social context.

Why it matters

- Case fatality is ~10% nationally, and higher for healthcare (16%) and senior-living (15%) exposures.
- Almost every case is hospitalized (~98%), so hospitalization tells us little. Death and length of stay are the useful measures of severity.
- Wisconsin now sees 200+ cases a year, and the January 2023 interview-form restructure gives three years of consistently structured WEDSS data.
- Findings feed DHS directly: which settings to prioritise for water-management follow-up.

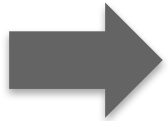
DATA UPDATE

What's new: DHS cleared sharing without PII, at census tract

After the Nov 3 DHS data-sharing call: you can share anything without personally identifiable information, and census tract is the resolution to use. Code sharing is fine. So: identifiers stay inside DHS, and a de-identified tract-level file comes to me.

INSIDE DHS (Rachel runs these — about 10 minutes)

1. **check_case_input(cases)**
   Aggregate QA: duplicates, unmatched counties/tracts, date gaps
2. **build_case_covariates(cases)**
   Joins county/tract SES, access, weather, storms (no network)
3. **make_analysis_file(cov)**
   Drops identifiers; turns dates into day-counts; random study ID
4. **county_rates(cases)**
   Crude, age-adjusted and SIR rates by county, for the map



LEAVES DHS

De-identified file: census tract, onset year and month, age (top-coded at 90), sex, race/ethnicity, conditions, exposure, outcomes as day-counts, area covariates

Aggregate county rate table (small cells flagged)

Never leaves: names, addresses, DOB, MRN, exact dates (unless DHS OKs onset date)

CODE UPDATE (V2)

Problems in the November code that would have bitten us, now fixed

Problem in v1 (Nov 3)	What would have happened	Fix in v2
Study window started 2024-01-01	All 2023 cases (about 1/3 of the cohort) get no weather or lag data	Window 2023-01-01 → 2025-12-31; weather from Nov 2022
Weather cache keyed only by year	Widening the window silently reuses a partial 2023 file	Cache named by exact dates and counties
Empty windows summarised as NaN / −Inf / 0	Fake zeros (“no rain, no storms”) enter the model	Returns NA; missing-day counter kept
Only YYYY-MM-DD dates, only FIPS codes	WEDSS/Excel exports (3/5/2024, county names) fail or go NA	Parses M/D/Y, Excel and SAS dates; county names → FIPS
Onset date required	Cases missing onset dropped from the lag analysis	Falls back to diagnosis → specimen → admit → report, shifted by the median gap and flagged
No input checks	Duplicate IDs or 2010-vintage tracts mis-join silently	check_case_input() prints a QA report
Output kept every column	Identifiers would ride along into any shared file	make_analysis_file() strips them

EXTERNAL COVARIATES

Public data attached to each case (county and census tract)

Layer (source)	Examples	Severity aim	Spatial aim
Social vulnerability (CDC SVI 2022; ACS with key)	poverty, uninsured, % 65+, race/ethnicity, housing	Confounder / effect modifier	●
Health status (CDC PLACES)	smoking, COPD, diabetes, CKD prevalence	Area proxy if WEDSS fields are sparse	●
NEW · Healthcare access (CMS, HRSA)	critical-access vs acute hospitals, primary-care HPSA, km to nearest hospital	● delay → severity	○
NEW · Population by age (Census PEP)	county × year × 5-yr age, 2020–2025	—	● age-adjusted rates and SIR
Rurality (USDA RUCC)	metro / non-metro	●	●
Weather (ERA5-Land) and storms (NOAA)	rain, humidity, heat in the 0–14 days before onset	○	● case-crossover
Water systems (EPA SDWIS) and facilities (EPA FRS)	surface water, cooling-tower proxies	○	●

● = main use ○ = secondary use. With tract cleared, exact tract values are kept; if DHS later prefers county, the code turns tract values into statewide quintiles.

VARIABLES

WEDSS fields to pull now (some aren't in the current plan)

Already planned

- Age, sex, race, ethnicity
- Smoking (current / former / never)
- Chronic lung, diabetes, CKD, CVD, cancer, immunocompromise
- Exposures: healthcare, senior living, travel, hot tub / recreational water, cooling towers
- Hospitalization, death, admit and discharge dates

Possibly missing: please check WEDSS

- **Interview:** Interview status, proxy vs self, interviewer — the main source of bias
- **Exposure detail:** CDC healthcare-associated category (presumptive ≥10 days / possible), not just “any healthcare”
- **Residence:** Lived in long-term care before onset
- **Diagnosis:** Urine antigen only vs PCR or culture. Urine antigen misses non-Lp1 strains, so test method can bias severity
- **Timing:** Onset, specimen, diagnosis and report dates (care-seeking delay)
- **Severity markers:** ICU, mechanical ventilation
- **Hospital:** Admitting hospital (or its type), transfers, discharge disposition (home / SNF)
- **Death:** Death date, and whether death was attributed to LD
- **Outbreak:** Outbreak- or cluster-associated flag
- **Immunosuppression type:** Transplant, chemotherapy, steroids, biologics (not one yes/no)
- **Geography:** Residence county, and tract if DHS geocodes it (2020 vintage)

ANALYSIS PLAN

Choosing the outcome: the “early death” problem

Death

~38–42 events (6%)

Clear and clinically meaningful. But ~10 events per parameter allows only ~4 terms, so use Firth logistic regression and a lean model.

Length of stay

~33% missing discharge dates

People who die on day 2 look like “mild” short stays. Naive LOS rewards dying early. Use only with death as a competing risk (Fine–Gray).

Hospital-free days

To day 30 · recommended primary

One ordinal score: 30 minus hospital days, with death scored worst. Uses every case and both outcomes. Proportional-odds model.

The template script (severity_analysis_template.R) runs all three on synthetic data, so we can compare them side by side when the real file arrives.

POWER CALCULATION

Power: death alone detects only large effects

Exposure prevalence	OR = 1.5	OR = 2	OR = 3
5% of cases (~32)	10%	26%	50%
10% of cases (~64)	14%	35%	69%
20% of cases (~127)	20%	44%	85%

Simulated power (500 runs each), unadjusted logistic regression, n = 635, 6% deaths, α = 0.05. Adjusting for confounders lowers these further.

What this means

- With death alone, a healthcare exposure that doubles the odds of death will usually be missed (~35% power).
- That favours an outcome that uses more of the data (hospital-free days), with death as a key secondary outcome reported with confidence intervals.
- Including 2023 (more cases) beats dropping it to get more complete outcomes, provided we impute or backfill the missing discharge dates.
- Next: rerun power_logistic() with the real exposure prevalences for the methods section.

DATA QUALITY

Missing data and bias: the decisions to make

Missing discharge dates

30–40% missing 2023–24; ~7% in 2025 (WISH)

Frances has already backfilled more 2024 dates. Open: is 2023 worth it? Missing stays are probably longer. Backfill if cheap; otherwise multiple imputation (ICU, ventilation, age, death; coded in the template). Report both.

Interview completeness

61% among deaths vs 85% overall

Keep “not interviewed” as its own exposure level. Sensitivity checks: interviewed cases only, imputed exposure, and a simple bias analysis.

Race/ethnicity and SES

Black residents ~19% of cases vs 6–7% of WI

A large disparity, so race is not just a nuisance confounder. Use individual race/ethnicity plus tract SVI, poverty and insurance; decide a priori what is confounder vs pathway. Draw the DAG.

2023 form restructure

start of a consistent data period

Check that exposure questions mean the same thing in 2023, 2024 and 2025. Include year in models.

SECONDARY AIM

Maps: adjust for population and age, and mask locations

Population-adjusted map (committee action item)

- LD risk climbs steeply with age, so a crude rate map mostly shows where older people live.
- county_rates() returns crude, age-adjusted (US 2000 standard) and SIR rates (observed / expected from statewide age-specific rates), with exact CIs.
- Empirical-Bayes smoothed SIR stops small counties from showing extreme rates on 1–2 cases.
- Cells below 5 are flagged for suppression. Confirm DHS's threshold.

Randomized-polygon / masked dot maps

- geomask_cases.R: each case gets a random point inside its county or tract, optionally weighted to populated census blocks.
- If DHS geocodes addresses, use donut masking instead: each point moves a random distance that scales with density, stays in its county, and hides among 50 or more residents.
- Resolution: census tract (DHS, Nov 2026). Posters and public maps may still need county or masked dots; confirm with DHS.

NEXT STEPS

To decide next week, and to do before the January meeting

Questions for DHS (Frances / Jordan)

- Exact onset dates: PII or not? tract + month + sex + age band will likely make nearly every case unique (k = 1)
- Small-cell threshold (<5 or <11) for tables and maps
- Is residence tract GEOID in WEDSS (2020 vintage), or only county/ZIP?
- Who owns the data dictionary and field-completeness table?
- 2023 discharge-date backfill: yes or no, and by when?
- Can we pull the fields on slide 6 (interview status, test method, ICU, disposition)?

Rachel, before late January

- Write the model as an equation: outcome ~ exposure + covariates
- Confounder list and DAG; mark which are measured
- Table 1 by outcome; cases-by-month plot (24 months); LOS histogram (is it bimodal?)
- Power with real exposure prevalences (power_logistic())
- Run check_case_input() and make_analysis_file() on the extract; send me the de-identified file
- Schedule the 2nd committee meeting (after Jan 20/21)

FILES

Where everything is

File	What it does
legionella_external_data.R (v2)	Step 1 pulls public data (no case data involved). Step 2 runs inside DHS: check → join → de-identify → rates
severity_analysis_template.R	Table 1, power, death model, competing risks, hospital-free days, imputation (runs on synthetic data)
geomask_cases.R	Randomized point-in-polygon and donut masking for maps
output/ (public covariates)	County and tract tables, weather 2022-11 → 2025-12, storms, hospitals, HPSA, population by age
CHANGES_v2.md	Line-by-line list of fixes and new variables

Google Drive: Rachel Erickson / Legionnaires / legionella_external_data / v2_2026-11-13 (CSVs in output/)